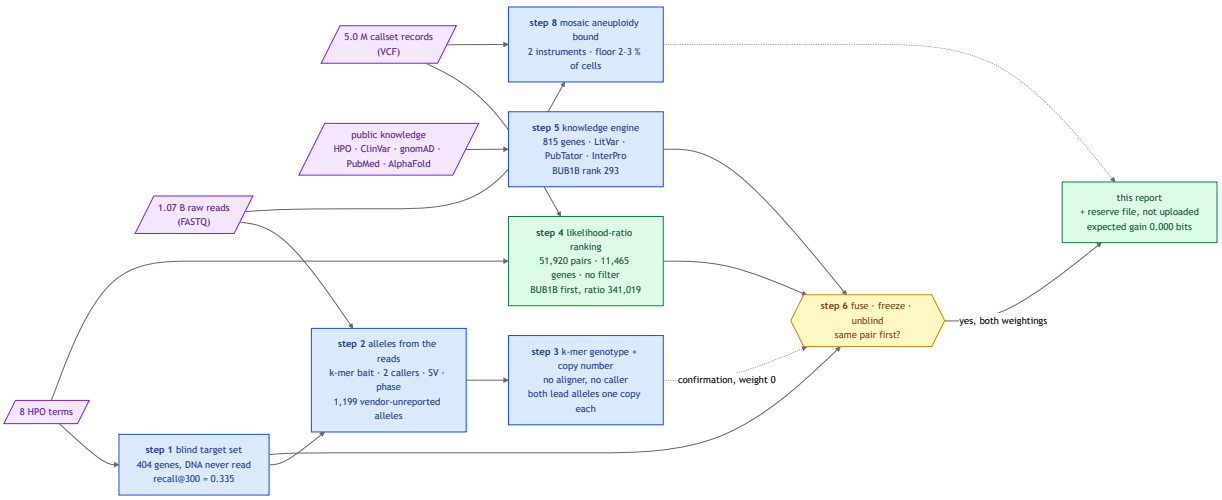
Track 1 report, iteration 2 — the same pair, found again with the answer hidden

Standing: the Track-1 replication report, written 2026-09-19 from executed runs. Every number below was produced by the methodology entry cited beside it (`docs/iteration2/step-<n>/methodology/`), and every claim is a pre-registered hypothesis in the append-only register (`science-register.json`, rendered at `figures/science-report.html`). Submission 1 was uploaded on 2026-09-09 and answered *full match at rank 1, 100 rank points, F-max 1.0* (083). **Submission 2 is built, rehearsed on two scorers and not uploaded**: it would teach nothing (124), so 5 of 6 uploads stay in reserve.

Written for a reader who knows software, not genetics. Each domain term is given as `technical term (plain meaning)` on first use; the full vocabulary is `dictionary.md`, served at `/learn` on the dashboard.

1. The answer in one paragraph

The child has **mosaic variegated aneuploidy (MVA)**: some of his cells carry the wrong number of chromosomes, because the checkpoint that holds a dividing cell until every chromosome is attached is weak. Iteration 1 found the cause top-down, from the vendor's variant calls: two different broken copies of **BUB1B**, the gene for that checkpoint's protein BubR1, one copy from each parent (a **compound heterozygote**) — a **stop-gain** allele (a premature full stop, already classified pathogenic for MVA syndrome 1) and a **missense** allele (one amino acid swapped) that no population database has seen. The challenge's answer key confirmed that pair at rank 1. Iteration 2 asked a harder question: **would we have found it without knowing it?** It started again from the raw reads, the callset and the eight clinical terms, under a **blind protocol** (no iteration-1 candidate, gene or table allowed as an input), by routes that share no code with iteration 1 — a symptom-only gene ranking, a k-mer read bait with two independent callers, an alignment-free genotyper, a genome-wide likelihood-ratio ranking with no hard filter, and a public-knowledge engine — fused them, froze the ranking, and only then looked. **The same two BUB1B alleles came out first, under both weightings** (109). The honest second half: one route found it, two corroborated it at a distance, one instrument confirmed it in the raw reads, and the literature never listed it. A second instrument then set the first real bound on the disease's defining feature: no chromosome in the sequenced sample carries a mosaic aneuploidy above 2–3 % of cells, where the earlier statistic had no power at all (116).



> **Legend** — **purple = raw input** (what the challenge delivered); **blue = machinery** (one research step each); **yellow = the decision point** (the unblinding); **green = the goal** (the one route with discriminating power, and the report). Dotted arrows are evidence that confirms but does not rank.

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2. What the reply said, and what iteration 2 is for

The scorer's reply to submission 1 — `full_match_rank 1`, `rank_points 100`, `f_max 1.0` at threshold 0.9963 — matched row 1 of iteration 1's pre-registered **decode table** (the lookup, built before the upload, that translates every possible reply into "which rows held the true alleles"). Survivors: `stop+missense` only. The seven alternative answer keys were eliminated in one upload; hypothesis H009 went to `supported` (083).

Nothing is left to *find* for the automated score. What is left is the question the automated score cannot ask: **is the pipeline right, or was it lucky?** Iteration 1 trusted the vendor's caller, applied hard filters (rarity, consequence) and checked that four instruments agreed. Each of those choices could have been the reason the pair surfaced. So iteration 2 is iteration 1's **mirror**, run **blind** (plan §1):

Allowed as input	Not allowed until step 6
the 8 HPO terms; the FASTQ; the VCF; public knowledge (HPO, Ensembl, gnomAD, ClinVar, PubMed, protein models)	the identity of the two iteration-1 alleles; the gene name as a filter; iteration 1's decode table, verdicts and figures
the step-0 platform (read store, warehouse, annotator, register, scorer replica)	any `results.*` table iteration 1 derived that encodes the candidate — iteration 2 re-derives its own under `results.it2_*`

The gene is written all over this repository, so the blinding is a **procedural rule, not a secret**: every methodology entry declares its inputs, and `tools/check_blind.py` (a CI gate) refuses an entry that names an iteration-1 candidate table before entry 109. Software analogy: not a sandbox, but a code review that rejects a forbidden import. The gate's current state: *38 iteration-2 entries, every one declaring its inputs — unblinded by 109*.

A pair rediscovered from scratch by a disjoint route is the strongest replication a single sample allows, and it is what makes the tooling transferable to a case with **no answer key**.

---

3. The clinical picture, and how both inputs were used

The challenge delivered eight **HPO terms** (Human Phenotype Ontology — a controlled vocabulary of clinical features, each with an id) and no karyotype:

Feature	HPO id	Where iteration 2 used it
Rhabdomyosarcoma (a childhood muscle-tissue cancer)	HP:0002859	step 1 ranking; step 5 case-report search; Track 2 direction D4
Nephrocalcinosis	HP:0000121	step 1; step 5
Short stature	HP:0004322	step 1; step 5
Failure to thrive	HP:0001508	step 1; step 5
Skeletal muscle atrophy	HP:0003202	step 1; step 5
Premature birth	HP:0001622	step 1; step 5
Small for gestational age	HP:0001518	step 1; step 5
Recurrent spontaneous abortion (**parental** history)	HP:0200067	step 1's `with-parental` query only; the recessive-with-lethality prior in step 4

Every step states which input it read. Step 1 read the clinical terms and **no DNA**. Steps 2 and 3 read the raw reads over a gene set that exists only because of the clinical terms. Step 4 read the VCF and the terms together, and its ablation says exactly what the terms added (§4.4). Step 5 read public knowledge against the seven proband terms. Step 8 read both raw inputs and the clinical document, and found in it no karyotype and no term a blood aneuploid fraction would predict (115). The warehouse's `data_lineage` table carries an edge from every derived table to both the delivered callset object and the clinical document by digest, and the submission's own three edges were appended by the resolver (124).

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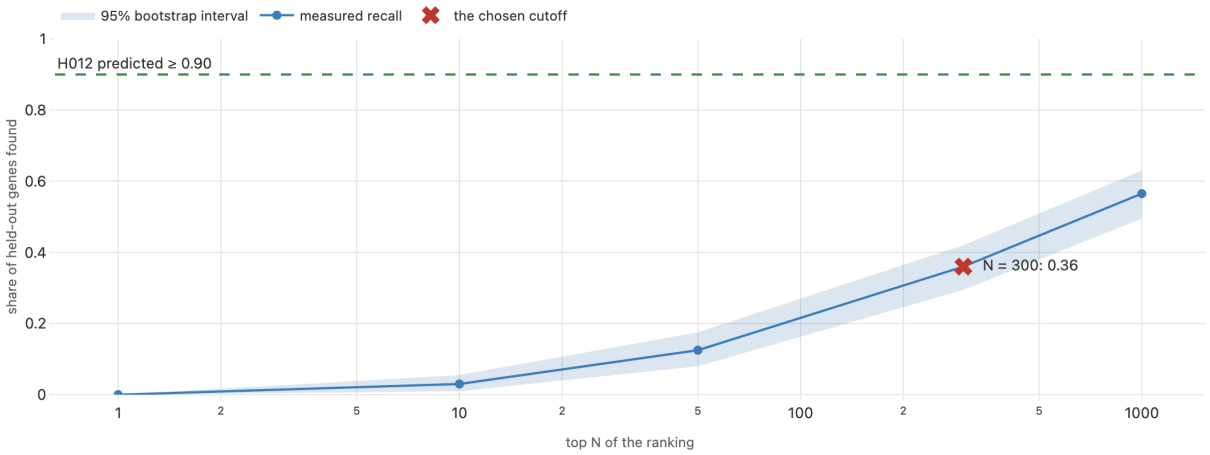
4. The mirror, step by step

Each step declared one goal in `src/mva/science/steps.py`, filed its hypotheses with a prediction and a refutation condition **before** running, and closed only when every hypothesis had a verdict. Refutations are reported with the same weight as supports: they are the half that changed the plan.

4.1 Step 1 — which genes do the symptoms alone nominate? (entries 084, 091)

A phenotype ranker (Exomiser's human-phenotype path, re-implemented in Python over HPO release 2026-06-23) scored 5,267 genes on the child's terms. Fixing N = 300 in advance would have been the usual move; step 1 **measured** it first, on a held-out benchmark of every recessive disease in the ontology with degraded query terms.

Measured	Value	Entry
target set	**404** genes in three tiers — `core` 62, `net` 272, `mechanism-term` 70 — plus **396** matched decoys (same length distribution, seed fixed)	084
held-out recall@300, full population of 3,240 diseases	**0.335** [0.318, 0.350]; recall@1000 0.508; **27.8 %** of held-out genes unrankable at any N	091
protein-coding genes with **no** HPO curation	**14,523 of 19,723 (73.6 %)**	084
agreement with a second ranker (Phen2Gene) in the top 300	**97 of 300 (32.3 %)**	084

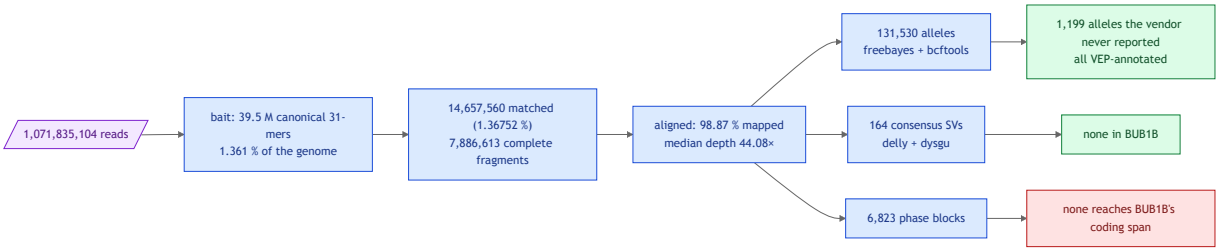


> **Legend** — **blue line** = measured recall with its 95 % bootstrap band, on the 200-disease sample of entry 084; **green dashed** = the 0.90 H012 predicted; **red cross** = the chosen N. The full-population number quoted above (0.335, entry 091) sits inside this band. An internal consistency check, not an external validation: ground truth and scoring database share curators.

Verdicts: **H012 refuted** (recall ≥ 0.90; measured 0.335), **H014 refuted** (rankers agree ≥ 50 %), **H015 refuted** (< 20 % uncurated; measured 73.6 %), **H013 supported** (a recessive chromosome-instability gene is in the set — 33 are; a sanity check, since the mechanism-term tier guarantees a seat). Consequence, recorded in plan §11: the target set is an **enrichment, never coverage**; "absent from the target set" excludes nothing; the genome-wide route (step 4) becomes the primary discriminator.

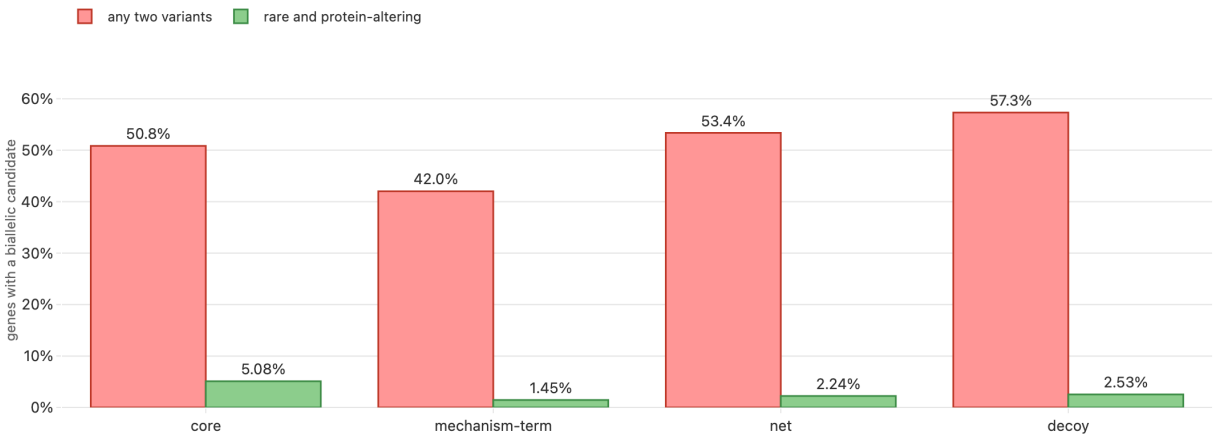
4.2 Step 2 — which alleles do the raw reads carry? (entries 085–090, 092)

The **read route** never touches the vendor's caller. A **k-mer bait** (every 31-letter window of the 792 target and decoy genes, packed into 2 bits per base and searched by binary search behind a bitmap prefilter — the new leaf package `mva.kmer`) selected from 1.07 billion reads the ones that touch those genes; those were aligned and called by two callers with disjoint models, and structural variants (large deletions, duplications, inversions) were called by two method classes.



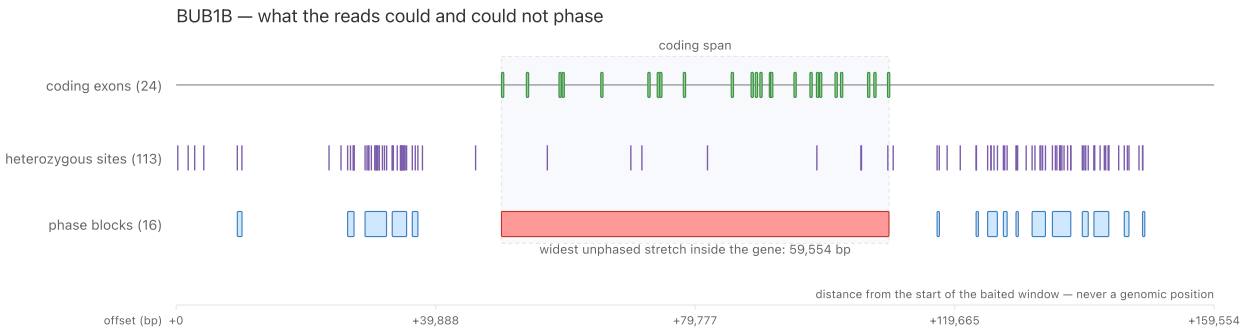
> **Legend** — **purple = raw input**; **blue = one stage each**; **green = what the route added**; **red = what it could not measure** (phase). Numbers from entries 087, 088, 090, 092.

Measured	Value	Entry
reads recovered against bait share of genome	**1.36752 %** vs **1.361 %** — the bait is not enriched for anything but its own size	087
bait recall on reads carrying a known allele	**0.9052** [0.815, 0.974]	087
concordance with the vendor's genotypes, substitutions only	`bcftools call` **0.9957** (45,316 / 45,512); freebayes **0.9927** (43,322 / 43,639)	088
alleles in or beside a coding exon the vendor never reported	**1,199**, all annotated	090
target vs decoy genes with a rare, protein-altering biallelic candidate, **novel alleles scored**	**10 / 396 vs 10 / 396**, Fisher one-sided p = **0.589**	090
the same, with **no** rarity or consequence filter	decoys **57.3 %** vs targets **51.0 %** — the decoys win	088
the lead pair's phase	**unmeasurable**: 10,911 bp apart, 4 heterozygous sites between, ~266 bp fragments, at 43.7×	088, 092
cost of the bait per read across a 36,000-fold bait-size range	**1.05×** — bound by bytes of sequence, not bait size	089



> **Legend** — **green = target genes**, **red = matched decoys**; left, no filter; right, the rarity + consequence filter with every read-derived allele scored. The bars are equal on the right: the symptom-derived list carries no signal of its own.

The finding that re-weighted the whole iteration. H020 predicted the target genes would carry biallelic candidates more often than random decoys. Its first reading (targets 4, decoys 2) was an **artefact**: the scoring query judged an allele by joining an annotation table built from the delivered callset, so an allele only the read callers saw could not be scored at all — a route built to find what the vendor missed could only rediscover what the vendor found. Scoring the 1,199 novel alleles too made the rates **exactly equal**. H020 is **refuted** on its pre-registered condition and its consequence applies: the read route enters the fusion as a **confirmation route at weight 0** — it says whether an allele is real, one copy, and not deleted; it never ranks. The discriminating power lives in the **filter**, not in the **gene list**.



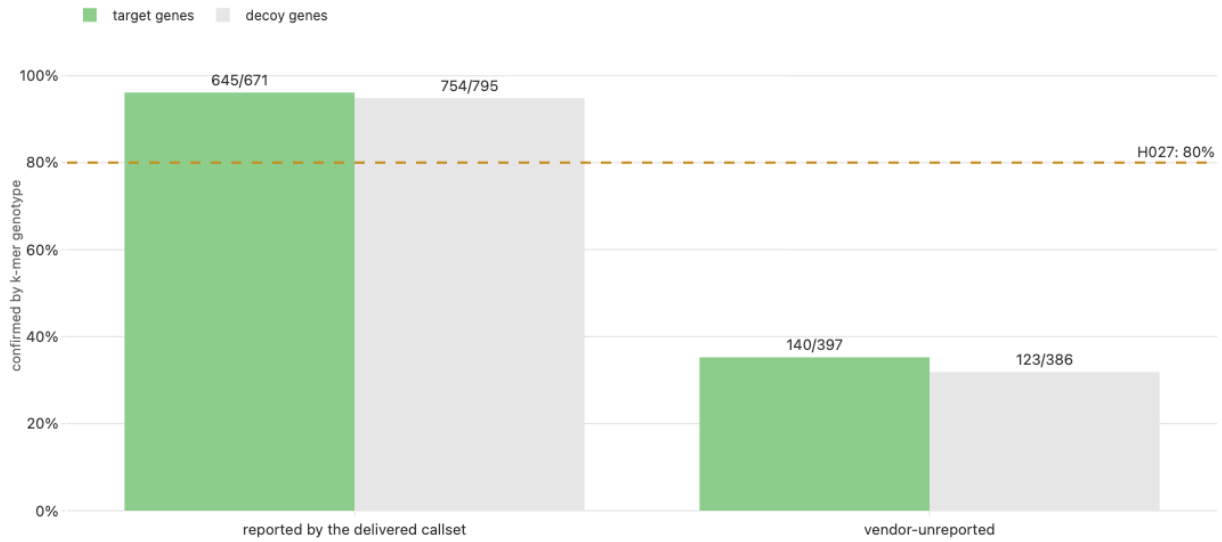
> **Legend** — the BUB1B exon rule with the heterozygous sites the reads carry and the read-backed **phase blocks** (stretches where the reads themselves say which alleles sit on the same chromosome copy). No block bridges the two lead alleles: **red = the unphased span**. This is a property of short-read fragments, not of depth; no deeper pass fixes it.

Verdicts: **H016, H019, H021, H022 supported**; **H020, H023 refuted** (two structural variants over an exon of a rare-allele gene were found, `NLRP7` net-tier and `OBSCN` decoy, both weak leads); **H017, H018 superseded** by H022/H023 because DeepVariant, Manta, LUMPY and GRIDSS have no arm64 build on this machine (089). The off-target bait gate (< 5 %) was **not reached** at any parameter — 8.79 % measured, 8.38–10.25 % across padding and k (086); it was mis-specified, and a per-gene bait-density gate replaced it.

4.3 Step 3 — genotype and copy number without an aligner (entries 093, 095–097)

A third instrument re-typed every allele the read route found — plus every ClinVar pathogenic allele in the target genes and 20,000 calibration markers — by counting the two 31-mers that differ only at the allele, straight from the baited reads: **no aligner, no caller**. A negative-binomial model (counts are over-dispersed; Poisson would be ~3× overconfident) fitted on the calibration markers turned counts into genotype posteriors.

Measured	Value	Entry
alleles typed / probe pairs kept	**164,044** / **381,510** of 439,851 designed (37,306 dropped because the alternate k-mer exists elsewhere in GRCh38)	095
fitted per-copy coverage	**17.96** → 35.93× diploid, against **35.2×** predicted from read length alone	096
the two lead alleles	both **heterozygous at posterior 1.0**, ref/alt counts **11/12** and **18/16**, 3 of 3 probes agreeing each	096
vendor-reported alleles confirmed vs vendor-unreported	**96.1 %** (645/671) vs **35.3 %** (140/397)	096
copy number, 1,430 regions	median **2.00** autosomal, **0.95** on X (the proband is male — the model was not told); null band **[1.028, 2.823]**; no BUB1B exon below it	097



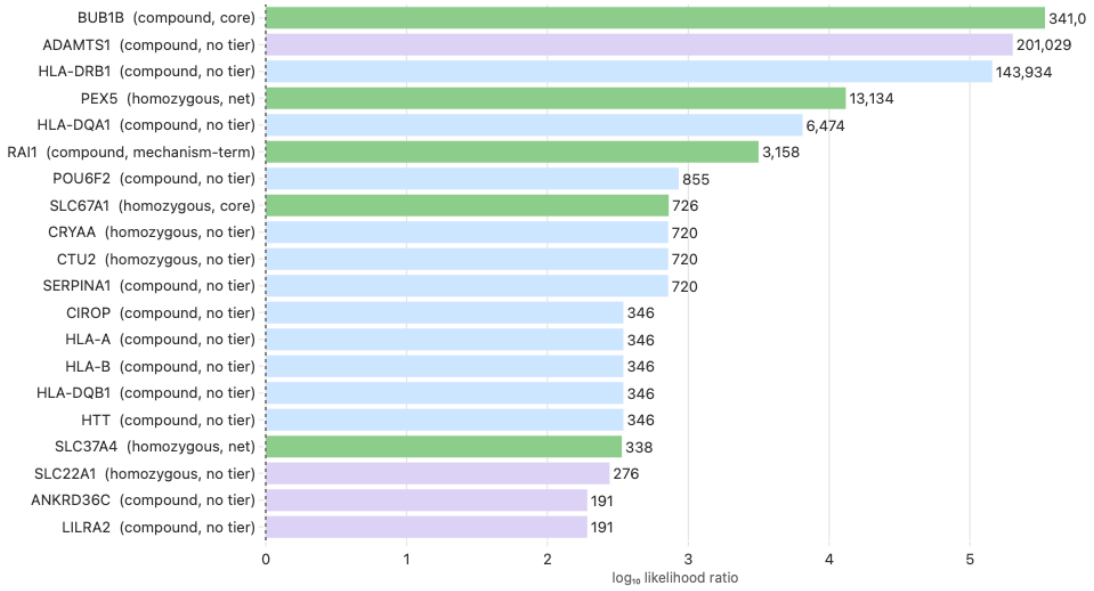
> **Legend** — **green = confirmed**, **red = contradicted**, **grey = no call**; by tier and by whether the vendor's caller had reported the allele. The 1,199 novel alleles are an **upper bound** with a measured error rate, not a caveat.

Verdicts: **H028 supported** (one copy each; 0.85), **H025 supported** (a ClinVar pathogenic allele present at heterozygous multiplicity — the BUB1B stop; four other "confirmed" ClinVar alleles are an artefact of an over-tight error rate and are said to be), **H024, H026, H027, H029 refuted**. H026's refutation surfaced one candidate autosomal exon loss unrelated to the pair — `BRD4` exon 7 at CN 0.85, net tier — flagged, not claimed. **One-copy dosage is not phase evidence**; the entry says so.

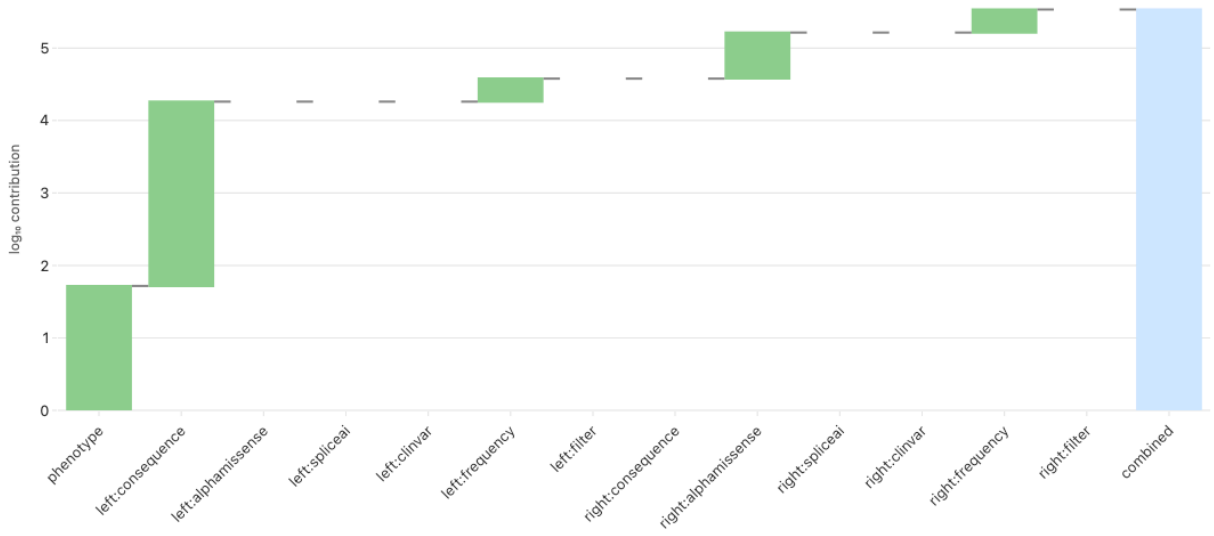
4.4 Step 4 — the genome-wide ranking with no hard filter (entries 098–101)

The one route with measured discriminating power. Every coding allele in the VCF — **all 38,087 of them, every FILTER value kept** — gets a **likelihood ratio** (how many times more likely the observation is if the allele is causal; a filter says in-or-out, a ratio says *×4.33*, and nothing is ever removed). One calibrated lane per consequence class (LOFTEE-style rules for truncating alleles, AlphaMissense for missense, SpliceAI for the splice window, ClinVar, gnomAD frequency, a non-PASS down-weight of ⅓) on the Tavtigian ACMG ladder; a phenotype lane calibrated on step 1's **own measured recall curve** (66.96× at rank 1 down to 0.379× for an unranked gene); a STRING network lane, capped at "supporting", for the 72 % of genes with no curation. Pairs, not genes, are the unit.

Measured	Value	Entry
biallelic hypotheses / genes	**51,920** over **11,465** — 35,120 compound, 14,130 homozygous, 2,670 single	099
leader	**BUB1B** stop + missense, ratio **341,019**	099
runner-up	`ADAMTS1` , ratio 201,029, **1.70×** behind — and phased **cis** by the caller's own physical phasing, so not a compound heterozygote	099
target vs matched-decoy separation	AUC **0.706** [0.655, 0.751]	100
permutation null (1,000 term sets matched to the child's)	**55 / 1,000** reach the observed leader, p = **0.056** — near-degenerate: only 4 genes in the callset can ever exceed it	100
DNA alone (phenotype lane silenced)	BUB1B **joint third**, ratio 6,548, behind `ADAMTS1` and `HLA-DRB1`	100
P(trans) from population co-occurrence	**undetermined** — 5 of 50 leading alleles reach gnomAD v2; reported at the uninformative 0.5	101



> **Legend — green = a step-1 target gene**, **red = a matched decoy**, **purple = an uncurated gene**, log scale. The leader stands 1.70× clear of a runner-up that is not a compound heterozygote at all.



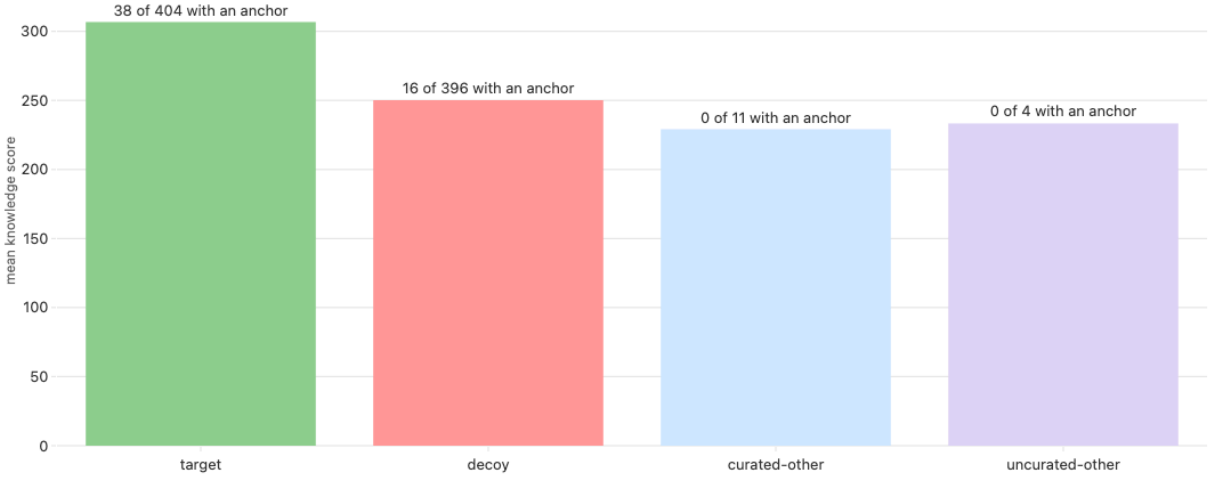
> **Legend** — the BUB1B pair's likelihood ratio decomposed into its lanes: **blue = genotype lanes**, **green = the phenotype lane**. The symptom route does the work of moving it from joint third to first.

Verdicts: **H030, H034, H035 supported**; **H031 refuted** (runner-up ≥ 10× behind: 1.70×), **H032 refuted** (an uncurated gene reached the top 20 — through the unranked-phenotype bucket, not the network lane), **H033 refuted** ($p < 0.001$ predicted; 0.056 measured). The step names its single most load-bearing assumption: the AlphaMissense and SpliceAI ACMG mappings could not be fetched and were **capped at *moderate***; lowering the cap to *supporting* flips the lead to `ADAMTS1` (098).

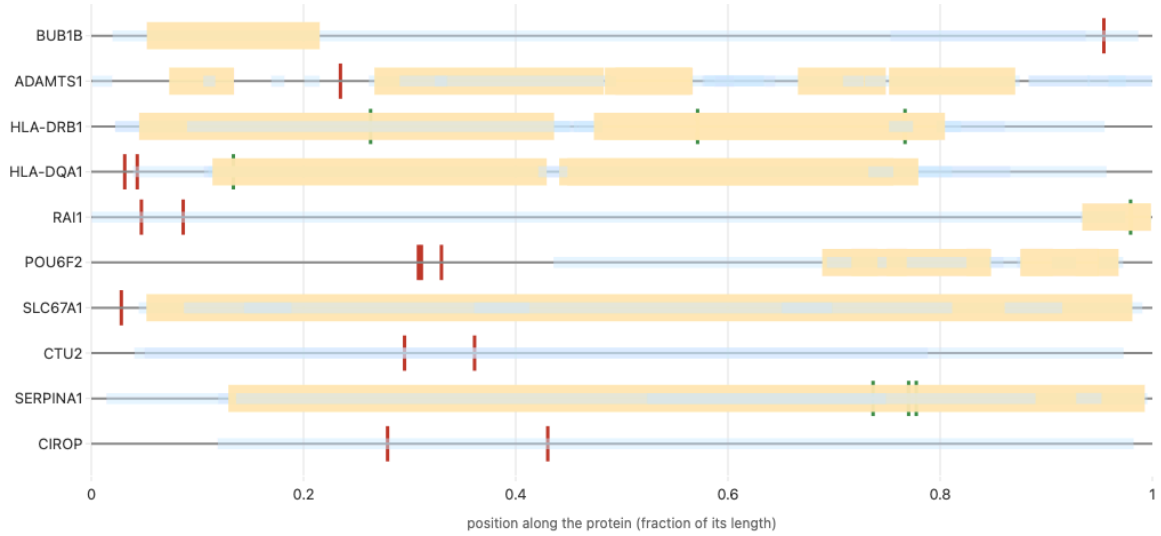
4.5 Step 5 — does public knowledge nominate a pair? (entries 102–106)

Against 815 genes (404 targets, 396 decoys, step 4's top 20): LitVar for published alleles, PubTator for papers co-mentioning a gene and each of the seven proband terms (5,705 searches), UniProt → InterPro → AlphaFold for where a missense sits in its protein, and PubMed hit counts as the **baseline the engine must beat**. Two planned items were cut and recorded: ESM1b (needs `torch`; AlphaMissense already covers every allele) and the LLM evidence tournament (the plan's own first cut).

Measured	Value	Entry
askable alleles with ≥ 1 publication	**1,821 of 1,914 (95.1 %)** — near saturation, orders almost nothing	103
ClinVar pathogenic alleles the child carries, typed in the raw reads	**15** across 14 genes, one of them the BUB1B stop	103
target vs matched-decoy separation	AUC **0.634** [0.593, 0.672] — below step 4's 0.706 on the same gene sets	105
against citation volume	Spearman ρ = **0.024**, top-5 overlap **0 of 5**	105
where BUB1B ranks by knowledge	**293 of 815**; the route's own top two genes are decoys, 10 of its top 25 are	105
the missense, in the protein	**95.4 %** along the chain, **outside every InterPro domain-typed entry**, AlphaFold pLDDT **91.06** (confidently modelled), AlphaMissense *likely_pathogenic*	106
best case report found blind for the leading gene	PMID 35804254 (MVA1 with a novel BUB1B variant), 3 of 7 terms — below H037's 5-term bar	104



> **Legend** — **green = targets**, **red = matched decoys**, the bootstrap band on the AUC. Length-matched decoys are essential: unmatched controls would measure gene size, not knowledge.



> **Legend** — **blue = InterPro domain-typed entries**, **purple = the family-level entry spanning the chain**, **red marker = the missense**, shading = AlphaFold confidence. The residue is past the kinase domain, in a confidently modelled stretch.

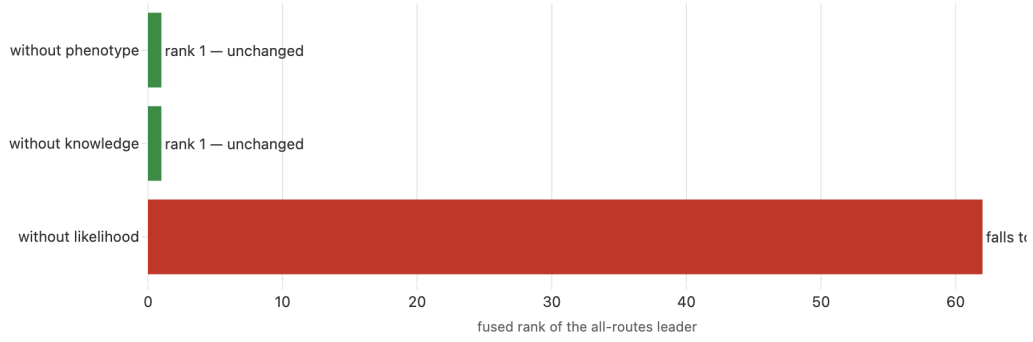
Verdicts: **H036, H038, H039 supported**; **H037 refuted** (a ≥ 5-term case report names a top-20 gene — the threshold selects surveys and guidelines, not case reports; no post-hoc fix was applied), **H040 refuted** (the missense is not in a domain-typed entry). The knowledge route is **corroborating, not discovery-grade**, and enters the fusion with that weight.

4.6 Step 6 — fuse, freeze, unblind (entries 107–109, 124)

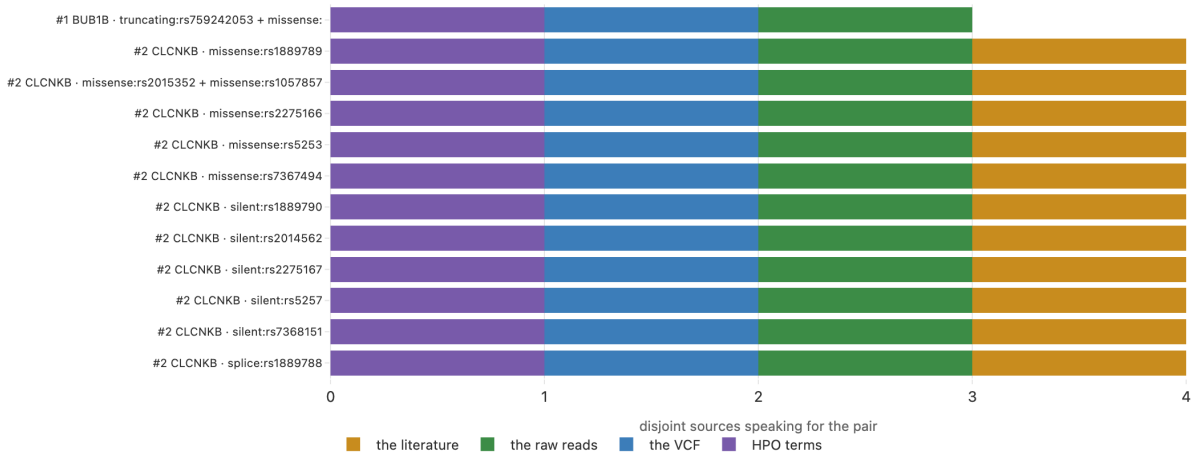
Three routes earned a vote by a rule fixed beforehand — a route ranks only if it separates targets from decoys **and** reorders a trivial baseline: `phenotype` (step 1), `likelihood` (step 4), `knowledge` (step 5). The read route (steps 2–3) confirms at **weight 0**. **Reciprocal rank fusion** (add `w / (k + rank)` from each list, k

= 60 fixed in advance, absent = rank 51, never 0) over each route's top 50 gave 274 fused pairs. The ranking was written and **hashed before the comparison** — sha256 `a596abf8…` (108). Only then did `tools/unblind_it2.py` read the pair to look for **from the committed iteration-1 selector file**, never by searching the ranking for a good answer.

Measured	Value	Entry
fused rank 1, equal weights	**the BUB1B stop + missense pair**	108
fused rank 1, measured-error-rate weights (phenotype 0.3346, likelihood 0.9441, knowledge 0.6337)	**the same pair**; `ADAMTS1` rises to 2, `CLCNKB` falls 2 → 49	108
the pair's rank on each route	likelihood **1** · phenotype **9** · knowledge **not listed**	109
leave-one-route-out	without `likelihood` → rank **62**; without `phenotype` → 1 (tied with decoy `DDX11`); without `knowledge` → 1	108
disjoint inputs behind the leader	**3 of 4** — HPO, VCF, reads; the literature is missing	108
what fills ranks 2–20	`CLCNKB`, 11 pairs (ranks 2–12); `DDX11`, a **decoy**, 8 pairs (13–20) — three genes, not twenty candidates	108
expected information gain of a second upload	**0.000 bits** on every expressible question	109, 124



> **Legend** — the leader's fused rank with each route removed. **Green = still first**; **red = falls to 62** when the likelihood route is silenced. The fusion was never a consensus.



> **Legend** — one segment per disjoint input (**HPO, VCF, reads, literature**); the BUB1B pair's literature segment is empty. `CLCNKB` scores 4 of 4 on this count and is still not the answer — a count of inputs is not a rank.

Verdicts: **H043 supported** (≥ 2 disjoint inputs; 0.80), **H044 supported** (a second upload is not informative; 0.90, re-tested on the built file in 124), **H041 refuted** (rank 1 *and* no route below rank 3 — two routes were), **H042 refuted** (no single route carries the result — the likelihood route does). Decision-table row 1 fires: **a replication report, and no upload**. The plan's fourth row — a second pair with comparable posterior — is *live in fact* (`ADAMTS1` , 1.70× behind) and *closed by the instrument*: the answer key holds one or two alleles, so "both" is not a reply this oracle can give.

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5. The two alleles, re-measured by iteration 2

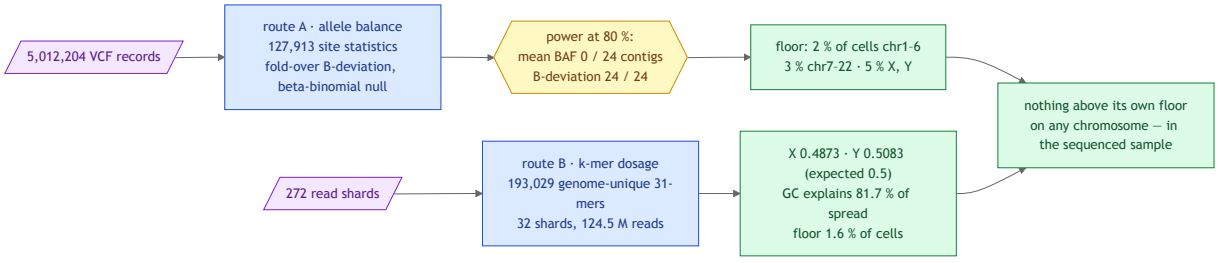
Iteration 1 characterised the pair from annotations. Iteration 2 measured it, by instruments that did not know which pair they were measuring. The Track 2 mechanism card (version 4) carries these as facts F1–F9; the primary sources are the Track 1 entries below.

Fact	Iteration 2's measurement	Entry
both alleles present, one copy each	heterozygous at posterior 1.0 in the raw reads, alignment-free; a **null + hypomorph** genotype, so some BubR1 is made	096
no hidden third allele	every BUB1B exon region at two copies; no consensus structural variant in the gene	097, 088
the missense is damaging	AlphaMissense *likely_pathogenic*, lane contribution 4.33 (capped at moderate); the pair leads 51,920 hypotheses with no filter	099
where the missense sits	past every domain-typed entry, 95.4 % along the chain, confidently modelled (pLDDT 91.06) — dosage, not folding, is the argued mechanism	106
phase (one allele per parental copy)	**unmeasured**: no read-backed phase block reaches the pair; population co-occurrence undetermined. *Trans* is the **oracle's** statement, not a read measurement	092, 101, 083
the literature	the gene ranks 293 of 815 by public knowledge; the blind script still found a citable MVA1 case report for it	105, 104

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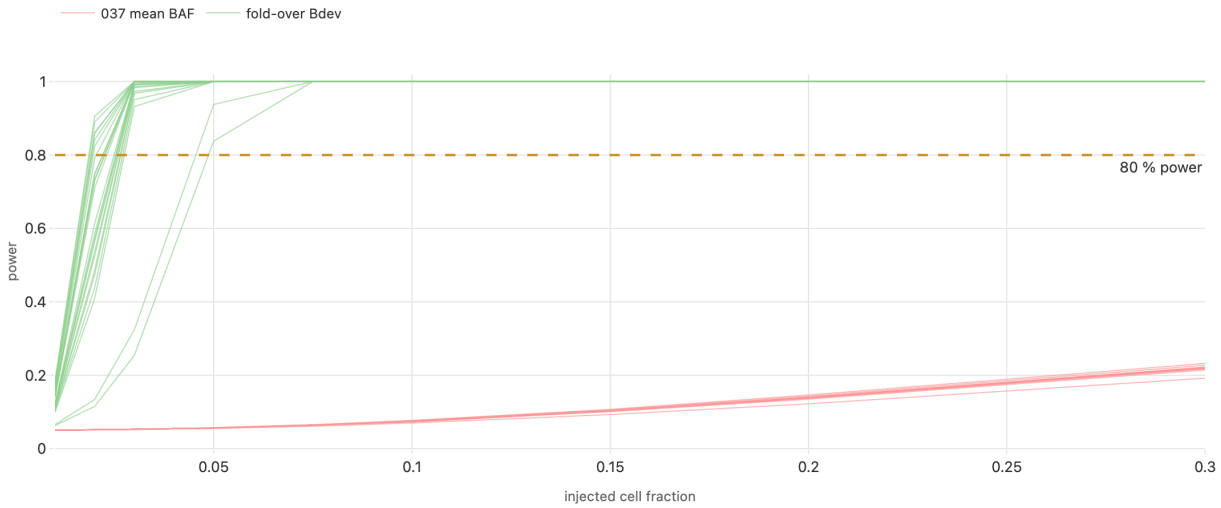
6. Step 8 — the disease's defining feature, bounded (entries 115–118, 121, 123)

MVA means some cells have the wrong chromosome count. Iteration 1's karyogram (entry 037) said every autosome looked flat. Step 8 asked what that was worth — and found the statistic had **no power**: a mosaic gain splits heterozygous sites into two bands symmetric about the centre, so their plain mean does not move. Folding the bands first (**B-deviation**, `|BAF − centre|` averaged *after* taking the distance) does.

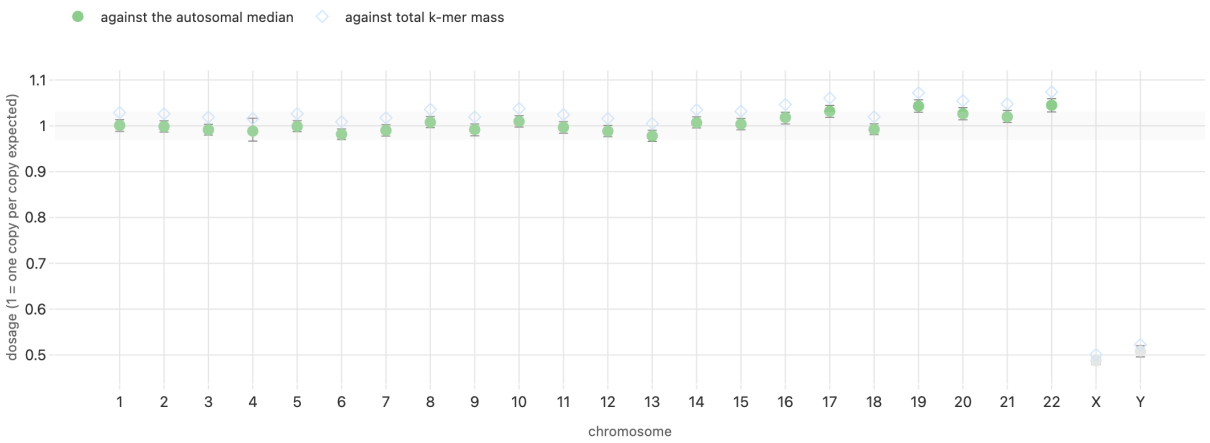


> **Legend — purple = the two raw inputs**; **blue = two instruments sharing nothing past the FASTQ**; **yellow = the power question**; **green = the bounds**. Route A is blind to X and Y in a male (almost no heterozygous sites); route B is blind to copy-neutral loss of heterozygosity. Where both can see, they must agree (H054).

Measured	Value	Entry
power at 80 %, 037's mean-BAF statistic vs fold-over B-deviation	**0 of 24** vs **24 of 24** contigs	116
detection floor, allele route	**2 %** of cells chr1–6, **3 %** chr7–22; nothing above its own floor	116
the divisor under a *variegated* genome	leave-one-out depth baseline **42.42×** [42.30, 42.64]; 1–3 injected abnormal chromosomes at 5–20 % recovered in **100 %** of 30 replicates, untouched ones false-flagged at **≤ 0.0048**	118
five-model mechanism comparison (Kong 2018, re-implemented)	**not usable**: 19 of 24 contigs end at a parameter bound; fitted fraction tracks the noise share at ρ = **0.938**	117
k-mer dosage, the positive control	X **0.4873** [0.4790, 0.4958], Y **0.5083** [0.4959, 0.5201] against 0.5 — the model was never told the sex	121
composition correction	chromosome GC explains **81.7 %** of the between-chromosome spread (1.876 % → 0.802 %)	121
chr21, the only autosome that moved	**4.03 σ** on probe GC → **2.33 σ** on the chromosome's own GC, Bonferroni p **0.43**; a mismapping pile-up tail (p99 depth 247 vs 66–69), not copy number	121
honest floor, dosage route	**1.6 % of cells** — twice the corrected residual spread, not the bootstrap width	121
the divisor that assumes nothing	absolute k-mer-mass and autosomal-median denominators agree on **24 of 24** contigs to **0.017 %**	123



> **Legend** — power against injected cell fraction, per contig: **red = entry 037's mean BAF**, which never reaches 80 %; **green = fold-over B-deviation**, which does on every contig.



> **Legend** — relative dosage per contig with bootstrap intervals; **blue = autosomes**, **green = X and Y at one copy**, the control to read first; the band is the composition-corrected null.

All seven claims supported (H051–H057). What this does **not** say: the tissue sequenced is not named by the delivery, a karyotype's fraction is measured in cultured cells, position was aggregated away so focal and sub-arm events are invisible by construction, and copy-neutral loss of heterozygosity is invisible to the dosage route. A bound on the sequenced sample, never a statement about the diagnosis.

---

7. The submission, read as a measurement (entry 124)

The scorer returns `full_match_rank`, `partial_match_rank`, `f_max` and its threshold. Read against the hypothesis space — every answer key still considered possible — a reply is a measurement. After the reply to submission 1 that space is **a single point**. The file iteration 2 would upload was nevertheless built from coordinate-free selectors, graded byte-for-byte on our replica **and** on the organisers' published `evaluation.py`, and decoded:

rank	epcr	type	row
1	0.9993	primary	**BUB1B stop_gained + missense** (H009)
2	0.4996	secondary	**LZTR1** heterozygous stop_gained, ClinVar P/LP — dominant schwannomatosis predisposition, unrelated to MVA
3	0.02498	secondary	**FLG** heterozygous stop_gained, ClinVar P — ichthyosis vulgaris / atopy
4	0.01499	secondary	**F5** Factor V Leiden, heterozygous — low-penetrance thrombosis risk

No number in this table was typed. The rank-1 epcr is the register's suggested confidence for H009 after the oracle's own reply entered its evidence (a transparent triage heuristic taken to its end, not a calibrated posterior); the secondary rows sit below it by allele score; the notes quote only columns of the table each allele was selected from (`derivation.md`). Iteration 1's alternative pairings are gone — the reply eliminated them.

Reply: full match rank	partial match rank	F-max	at epcr	prior mass	survivors
1	—	1.000	0.9993	1.00	stop+missense

Both scorers agree: **100 rank points, F-max 1.0**. Prior entropy **0 bits**, expected information gain **0.000 bits**. The plan's rule (§6) is that an upload is spent only when a pre-registered decode table shows a gain above zero on a live hypothesis. It does not. **Not uploaded; 5 of 6 remain, deliberately.** The file is

`submissions/track1/iteration-2/agharsallah_bub1b-comphet-replication.csv` (in the iteration folder, gitignored; sha256 `325205a8…`; rebuild in `README.md`). Timing of any upload is the team's call.

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8. Reproducibility, runtime and cost

Every step has a `How to reproduce it` block in its README under `docs/iteration2/`; the commands below are the spine.

```
 # step 1 — the blind target set (laptop; the full benchmark ~2 h 16 min on one core, entry 09
uv run python tools/build_it2_targets.py && uv run mva-phenotype benchmark --sample 3240 --see
# step 2 — bait, one pass over 1.07 B reads (server-side POST /bait, 50.8 min; 4.2 h projected
uv run python tools/build_it2_bait.py && uv run python tools/run_it2_step2_calling.py
# step 3 — k-mer genotypes and copy number from the baited reads (laptop; 765 s + 512 s, entri
uv run python tools/build_it2_probes.py && uv run python tools/run_it2_step3_counts.py && uv r
# step 4 — likelihood-ratio ranking (laptop; the 1,000-draw permutation null 34 min, entry 100
uv run python tools/rank_it2_pairs.py
# step 5 — the knowledge engine (free public APIs; 5,705 PubTator searches, entry 104)
uv run python tools/fetch_it2_catalogue.py && uv run python tools/build_it2_anchors.py --inter
# step 6 — fuse, freeze, unblind
uv run python tools/fuse_it2_routes.py && uv run python tools/assess_it2_upload.py && uv run p
# step 8 — mosaic bound (route A laptop; route B 32 of 272 shards server-side, 2,803.8 s, entr
uv run python tools/fit_it2_mosaic.py --stages constants bdev baseline models limits multi
uv run python tools/run_it2_dosage_pass.py --upload --subsample 32 --workers 8
# the submission
uv run python tools/resolve_submission_spec.py --selectors submissions/track1/iteration-2/spec
uv run python tools/design_submission.py --spec build/submission/spec.json --out submissions/t
```

Hardware: one arm64 laptop (16 GiB) for everything but two read passes, which ran on a small Linux server holding the 272-shard read store; a hosted PostgreSQL for the warehouse; free public APIs (Ensembl VEP REST, LitVar, PubTator, UniProt, InterPro, AlphaFold DB, PubMed, STRING). **No paid compute.** The k-mer engine's cost is bound by bytes of sequence, not by bait size (1.05× per read across 36,000× of bait), which is what makes "give me the reads for these genes" a case-agnostic tool. What the arm64 machine could not run is recorded, not hidden: DeepVariant, Manta, LUMPY, GRIDSS, Exomiser/LIRICAL — substituted by freebayes + `bcftools call`, delly + dysgu, and step 1's own ranker.

The tooling is generic: a reference k-mer read bait and counting kernel (`mva.kmer`), a negative-binomial k-mer genotyper and local-flank copy number, a per-lane calibrated likelihood-ratio pair ranker, a reciprocal-rank fuser with leave-one-out and a freeze receipt, a knowledge fetcher with matched-decoy baselines, a mosaic-fraction bound with two instruments and an injection harness, a blind gate, and a per-iteration step map. Every table is in `warehouse.data_lineage`; every figure is drawn from a run's own output file. `tools/check-python.sh` runs the eight quality gates, including the coordinate gate and the iteration-folder gate.

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9. Limitations, stated

- **One route carried the rediscovery.** Silence the likelihood ranker and the pair falls to rank 62; the symptom route had it at rank 9; the literature never listed it. "Three routes agree" would be false; "one found it, two corroborated it at a distance, one confirmed it in the reads" is the sentence (109).
- **The rank-1 position depends on the clinical terms and on one calibration cap.** DNA alone puts the pair joint third; capping the missense and splice lanes at *supporting* instead of *moderate* flips the lead to `ADAMTS1` (099, 100).

- **Phase is unmeasured.** No read-backed phase block reaches the pair and population co-occurrence is undetermined. *Trans* rests on the recessive model and on the oracle's answer key; iteration 2's own instruments cannot reach it.
- **A rediscovery is not coverage.** The symptom ranker's held-out recall@300 is 0.335, and 73.6 % of protein-coding genes have no HPO curation; the route found *this* pair and says nothing about the next case. The genome-wide ranking is the safety net, and its permutation p is 0.056, not < 0.001.
- **The read route has no ranking power of its own** (H020 refuted); its 1,199 novel alleles are corroborated at 35.3 % by the alignment-free instrument and are an upper bound with a measured error rate. Concordance with the vendor covers substitutions only; the structural variant consensus used two of four planned callers and no positive-control sample.
- **Below rank 1 the fused ranking has poor resolution.** Rank inheritance makes the top 20 three genes; a matched decoy holds ranks 13–20. The confirmation triple is gene-level, not allele-level.
- **The literature route is weak by measurement**, AUC 0.634, and would not have found the gene (rank 293 of 815). A publication is not evidence about this child; 95.1 % of alleles have one.
- **The mosaic bound is about the sequenced sample.** Tissue unstated; focal and sub-arm events invisible by construction; copy-neutral loss of heterozygosity invisible to the dosage route; the mechanism-of-origin models do not fit this data at all. The dosage route's divisors were checked against each other (0.017 %) but were not put through the injection harness the allele route was (118, 123).
- **Confidence numbers are a heuristic.** The register's weighing is transparent arithmetic for triage, not a calibrated posterior; the epcr of 0.9993 inherits that caveat.
- **Some of the plan's instruments never ran on this hardware**, and two exhibits (the per-exon depth grid, the BUB1B phase track) are committed figures not yet on the dashboard route (`docs/iteration2/step-2/notes/completion-audit.md`, D7–D8).

10. Methods form (Track 1), model 2 — answered

Question	Answer
Model	model 2: the **blind mirror** — target set → k-mer read bait + two callers → alignment-free genotype and copy number → genome-wide likelihood-ratio pair ranking → knowledge engine → reciprocal-rank fusion, frozen, then unblinded
Automated or curated?	**Automated** to the fused ranking and its rank-1 pair, which was hashed before anyone compared it to iteration 1. The submission layout (one primary pair, three secondary rows) follows a stated rule and was simulated; nothing in it is a belief typed by hand
Public data only?	**Yes** — Ensembl 116 and VEP REST (gnomAD v4, ClinVar, AlphaMissense, SpliceAI), HPO release 2026-06-23, STRING, LitVar, PubTator, UniProt, InterPro, AlphaFold DB, PubMed, GRCh38
Pairs or singles?	**Pairs**: every hypothesis in step 4 and every fused unit in step 6 is a biallelic pair; the primary row holds both alleles
Secondary findings	ClinVar P/LP after excluding conflicting and benign-tagged records, gnomAD AF < 0.02, dominant or risk-factor genes → LZTR1, FLG, F5 as `secondary`; recessive carrier states reported, not submitted
Runtime and cost	two read passes on a small Linux server (50.8 min; 2,803.8 s for 32 of 272 shards), the rest on one laptop; the longest laptop jobs are the 3,240-disease benchmark (~2 h 16 min) and the permutation null (34 min); free public APIs; one small hosted PostgreSQL; no paid compute
Abstract	`methods-form.json`, question 18, under 500 words; strengths and limitations from §1 and §9

11. Where every claim lives

Step	Supported	Refuted or superseded	Entries
1 target set	H013 (0.83)	H012 (0.04), H014 (0.10), H015 (0.02)	084, 091
2 reads	H016 (0.88), H019 (0.70), H021 (0.65), H022 (0.88)	H020 (0.07), H023 (0.25); H017, H018 superseded	085–090, 092
3 k-mer genotype	H025 (0.60), H028 (0.85)	H024 (0.15), H026 (0.30), H027 (0.08), H029 (0.20)	093, 095–097
4 likelihood ratio	H030 (0.75), H034 (0.95), H035 (0.86)	H031 (0.14), H032 (0.25), H033 (0.15)	098–101
5 knowledge	H036 (0.80), H038 (0.78), H039 (0.85)	H037 (0.15), H040 (0.20)	102–106
6 fusion	H043 (0.80), H044 (0.90)	H041 (0.20), H042 (0.12)	107–109, 124
8 mosaic bound	H051 (0.80), H052 (0.85), H053 (0.95), H054 (0.85), H055 (0.80), H056 (0.85), H057 (0.85)	—	115–118, 121, 123
the pair (iteration 1)	H009 (supported; oracle)	—	074, 083

Confidences are the register's committed values (`mva-science board`); where a committed value sits below the arithmetic's suggestion, the hypothesis file records why. Step 7 is Track 2's step (H045–H050, H061, all supported) and is reported in `submissions/track2/iteration-2/`, a draft built on the confirmed pair and on the facts above; it does not test the pair and is not cited here as evidence for it.